Flexible behavior or flexible methods? A cross-taxon review of experimental designs in reversal learning

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Competing Interests

All authors are members of the ManyManys collaboration, which is referenced in the manuscript. The authors declare no competing interests.

Data Availability Statement

The complete dataset, analysis script, and figures are available on Figshare. A public link will be made available upon acceptance.

Abstract

Behavioral flexibility—the ability to adapt behavior in response to changing conditions—is widely recognized as a key feature of animal cognition. It is often measured using reversal learning tasks, where individuals must inhibit a previously rewarded response and adopt a new one after contingencies shift. Despite its widespread use, the comparability of these tasks across species remains unclear. This paper establishes a foundation for resolving this issue by examining how reversal learning has been designed and implemented across taxa. We conducted a systematic review of 206 empirical studies (2014–2023) spanning eight major taxonomic groups: invertebrates, fishes, amphibians and reptiles, birds, rodents, non-human primates, humans, and other mammals. For each study, we extracted variables related to taxon coverage, sampling, learning and reversal criteria, cue types, and outcome measures. Analyses included nonparametric tests to assess group-level differences, linear discriminant analyses to explore multivariate structure, and model-based robustness checks. We identified three methodological obstacles to understanding reversal learning across diverse taxa. First, the distribution of research is highly imbalanced: birds, rodents, and humans accounted for most of the populations in the reviewed work. When considering species coverage, most animal diversity—especially invertebrates, fishes, and amphibians and reptiles—remains virtually untested, with less than 1% of described species included per taxon. Second, research is taxonomically fragmented: 99% of studies focus on a single group, limiting opportunities for direct comparison. Third, and most critically, methodological standards diverge dramatically across taxa. Humans are consistently held to the strictest learning criteria, while other taxa most often use lower thresholds. The number of reversal phases differ more than threefold among taxa. Nearly all studies of amphibians and reptiles, fishes, and invertebrates use single-reversal designs, whereas multi-reversal protocols are much more common in humans and non-human primates. Sample sizes, evaluation window lengths, cue types, and outcome metrics also display taxon-specific patterns. These systematic differences in experimental design introduce structural asymmetries that complicate cross-taxon comparisons, blurring the line between true cognitive variation and methodological artifacts. Although research to date has advanced our understanding, further progress will depend on greater methodological coordination and broader taxonomic coverage. Emerging large-scale collaborations are beginning to address these gaps, offering a promising path toward a more robust and equitable science of behavioral flexibility.

Keywords: behavioral flexibility, reversal learning, cross-species study design, task comparability, design standardization, comparative cognition.

Flexible Behavior or Flexible Methods? A Cross-Taxon Review of Experimental Designs in Reversal Learning

The capacity to flexibly modify behavior in response to changing environmental conditions enables organisms to exploit new resources, avoid threats, innovate solutions, and navigate complex social dynamics. These skills are essential for increasing overall fitness, specifically in terms of survival and reproductive success (Izquierdo et al., 2017; Kringelbach, 2004). In the context of the Anthropocene, an era marked by accelerated, human-driven environmental disruption, understanding behavioral flexibility has taken on new urgency. Species that can flexibly adapt to altered habitats, changes in climate, and novel anthropogenic pressures may be more likely to survive, making behavioral flexibility a key target for quantifying and predicting their extinction risk and adaptive potential (Ghisbain et al., 2025; Mazza & Šlipogor, 2024; Vinton et al., 2022).

Reversal learning is a widely used paradigm for assessing behavioral flexibility across species—including humans (Bond et al., 2007; Harlow, 1944, 1949). In its basic form (Figure 1), an individual is first trained to discriminate between two stimuli, one of which is consistently associated with a reward. After the individual meets a predetermined learning criterion—typically defined as a percentage of correct responses over a series of trials—the stimulus-reward contingencies are switched: the stimulus that was previously rewarded no longer leads to a reward, and the previously unrewarded stimulus becomes the correct choice. Successful performance after the reversal requires inhibiting the previously reinforced response and selecting the newly rewarded stimulus. Species differ markedly in how they respond to this shift. Some struggle to acquire the new contingencies, while others adapt more efficiently over time. To capture such patterns, some studies implement repeated reversals, allowing researchers to examine changes in performance across successive contingency shifts—a process referred to as *interreversal learning* or *serial reversal learning* (Mackintosh et al., 1968;

Schusterman, 1966; Warren, 1960). This approach provides insight into how individuals generalize across similar but non-identical situations—so-called *learning sets* (Harlow, 1949)—and offers a window into the emergence of meta-learning, or the capacity of "learning to learn", a presumed hallmark of cognitive development (Adolph, 2019).

Reversal learning is widely regarded as a gold-standard paradigm for evaluating behavioral flexibility across species. Two key features make it especially well suited to comparative cognition. First, it effectively captures the environmental instability that animals encounter in their daily lives—such as spatiotemporal changes in food location in frugivores (Milton, 1981) or shifting social contexts in group-living species (Amici et al., 2018). Second, reversal learning supports extensive experimental customization. For example, researchers can tailor the task to species-specific sensory modalities and ecological constraints by incorporating visual (Gaalema, 2011; Salwiczek et al., 2012), visuospatial (Daniel & Schluessel, 2020), olfactory/gustatory (Brushfield et al., 2008; Pérez Claudio et al., 2018; Scarlet et al., 2009), auditory (McMillan et al., 2017), or haptic cues (Bonney & Wynne, 2002). Similarly, they can adjust reinforcement types, contingency structures, and reversal frequency (Erlebacher, 1963; Kendler & Kimm, 1964; Strang & Sherry, 2014). This level of control allows researchers to optimize task design to suit the perceptual, motivational, and cognitive profiles of different species.

Reversal learning has become one of the most broadly applied paradigms in comparative cognition, with extensive implementations across an exceptionally wide range of taxa, such as arachnids (e.g., *Marpissa muscosa*: Liedtke & Schneider, 2014; *Aphonopelma hentzi*: Punzo, 2002), cephalopods (e.g., *Octopus bimaculoides*: Boal et al., 2000; *Octopus vulgaris*: Bublitz et al., 2017; *Sepia officinalis*: Karson et al., 2003), insects (e.g., *Apis mellifera*: Chandra et al., 2000; *Nauphoeta cinerea*: Longo, 1964; *Bombus terrestris*: Raine & Chittka, 2012; Heliconiini butterflies: Young et al., 2024;), fishes (e.g., *Poecilia reticulata*: Lucon-Xiccato & Bisazza, 2014; *Pseudochromis aldabraensis*, *P. flavivertex*, *P. fridmani*: Prétôt et al., 2025; *Labroides dimidiatus*: Salwiczek et al., 2012), amphibians (e.g., *Triturus viridescens*: Ellins et al., 1982; *Dendrobates auratus*: Liu et al., 2016), reptiles (e.g., *Varanus rudicollis*: Gaalema, 2011; for a review of non-avian reptiles, see Szabo et al., 2021), birds (e.g., *Parus major*: Cauchoix et al., 2017; *Phasianus colchicus*: Madden et al., 2018), rodents (e.g., *Rattus norvegicus*: Galizio et al., 2023; *Mus musculus*: Le et al., 2023), marine mammals (e.g., *Tursiops truncatus*, *Zalophus californianus*: Beach III et al., 1974; *Phoca vitulina*: Niesterok et al., 2022), ungulates (e.g., *Capra hircus*: Langbein, 2012; *Bos taurus*: Meagher et al., 2015), carnivores (e.g., *Canis latrans*, *Mephitis mephitis*, *Procyon lotor*: Stanton et al., 2021; for a review of mammalian carnivores, see Benson-Amram et al., 2023), non-human primates (e.g., *Pan troglodytes*: Cantwell et al., 2022; *Cebus apella*, *Macaca silenus*, *Saimiri sciureus*: Judge et al., 2011; *Gorilla gorilla gorilla*, *Mandrillus leucophaeus*, *Pongo* spp.: Prétôt et al., 2021), and humans (*Homo sapiens*: Clark et al., 2004; Postalli et al., 2015). Researchers have frequently reported success in detecting reversal learning across this broad taxonomic range, with animals demonstrating the ability to inhibit previously learned associations and acquire new ones—findings that are widely interpreted as evidence of behavioral flexibility.

Despite its widespread use, reversal learning is often implemented in ways that are both methodologically variable and taxonomically narrow. For instance, most studies concentrate on a single—or at most a few—species, relying on procedures designed for specific experimental settings and available resources. An extreme case of species specialization is found in neuroscience studies, which have traditionally been conducted with mice (Bissonette et al., 2008), rats (Ragozzino, 2007; Schoenbaum et al., 2000), rhesus macaques (Clarke et al., 2004) and, more recently, common marmosets (Takemoto et al., 2015). In addition, relatively little is known about how specific design choices that can be rather different across studies—such as cue types, learning criteria, or overtraining—affect performance. Cue types, for example, vary widely across studies, ranging from simple visual features to spatial, olfactory, or auditory signals, introducing major differences in perceptual demands. Other parameters, such as learning criteria and overtraining—the continued reinforcement of a correct response after the learning criterion has been reached—also differ substantially. These variations reflect the versatility of the reversal learning paradigm but also lead to methodological heterogeneity that limits the comparability of different studies and poses serious challenges for interpreting differences—especially when species differ in anatomy, sensory systems, motor constraints, or ecological backgrounds (Alessandroni et al., 2024, 2025).

Before the field of comparative cognition can meaningfully compare behavioral flexibility across species, it must first gain clarity on how reversal learning research has actually been conducted. Without understanding the empirical practices that inform this literature, we cannot know whether cross-species findings reflect true cognitive differences or methodological variation. To provide this essential foundation, we conducted a systematic review across a broad range of animal taxa. Critically, our goal in the current paper was not to compare reversal learning performance between species, given the challenges posed by methodological heterogeneity in the field—rather, we systematically mapped how these studies have been designed and implemented, documenting the diversity of procedural approaches, task parameters, and measurement practices. This systematic review of the field's methodological landscape is necessary for establishing standardized, interpretable approaches to studying behavioral flexibility in comparative cognition.

Method

Scope and Aims of the Review

Using the Web of Science (Clarivate ®) database, we searched for original empirical research articles published in English between January 1, 2014, and September 20, 2023. This window was selected to capture current experimental practice while constraining the number of reviewed papers to a manageable size.

Our goal was to characterize methodological diversity and identify structural features of task design that may shape how behavioral flexibility is measured across species (see Table 1 for a summary of key research questions). We considered dimensions related to taxonomic representation (e.g., which groups are most frequently studied, how species are distributed within taxa, and whether studies compare multiple species or focus on a single one), as well as features of sampling and evaluation—such as sample size, the number of trials used to assess learning and reversal, and the strictness of performance criteria. We also examined core elements of task design, including whether overtraining is implemented, the number of reversal phases used, and the types of cues presented. Finally, we explored whether certain outcome measures are reported more frequently for specific taxa.

Search Strategy and Inclusion Criteria

The full search query was: TS=((animal* OR avian OR bird* OR rodent* OR mouse* OR mice OR rat* OR carnivore* OR primate* OR monkey* OR ape* OR mammal* OR dog* OR canid* OR cat* OR goat* OR sheep* OR bovidae* OR horse* OR insect* OR ant* OR bee* OR spider* OR amphibian* OR frog* OR gupp* OR reptile* OR fish* OR cephalopod* OR octopus* OR human* OR adolescent* OR adult* OR child* OR infant* OR bab*) AND “reversal learning”) NOT TS=(atypical OR mutant* OR genetic* OR disorder* OR *degeneration OR disease* OR dysfunction* OR antidepressant* OR impair* OR diminish* OR chromosom* OR deficit OR drug* OR injur*).

The literature review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). The search initially returned 350 articles. To be included, studies had to (i) focus on reversal learning, (ii) involve at least one taxon of interest as defined by the search terms, and (iii) describe an experimental procedure. A total of 208 articles met these criteria, while 142 did not and were thus excluded. Reasons for exclusion included non-ecologically valid experimental conditions—such as pharmacological interventions or genetically modified models (*n* = 73); articles focused primarily on neural activity without accompanying behavioral measures—for example, neuroimaging or electrophysiological studies (*n* = 27); studies that, upon examination, were found not to implement a reversal learning procedure despite their appearance in database results (*n* = 22); and non-empirical publication types (e.g., reviews or meta-analyses) (*n* = 13). An additional seven articles were classified under "other" indicating exclusion for reasons not covered by the main categories, such as atypical or insufficiently described reversal learning procedures and articles reporting previously published data. A full overview of the selection process is provided in Figure 2.

Data Coding

All included articles (*n* = 208) were reviewed using a predefined coding scheme developed specifically for this project. Two rounds of pilot coding were conducted by subgroups of the author team to refine category definitions and ensure coding consistency. We define a *cell* as a unique group of subjects from a given species tested under a specific experimental design, and *study* as a single published article. Articles involving more than one species or experiment were coded as multiple cells. If at least one cell met the inclusion criteria, the article was retained, and only its eligible cells were included in the analysis. The final dataset comprised 271 such cells.

To assess consistency, a subset of 75 articles was independently double-coded. All 75 were included in the comparison of decisions about whether to retain the article for the review. Agreement on these inclusion decisions was very good (Gwet's AC1 = 0.96), following Altman's thresholds (Altman, 1991). Of the 72 articles both coders independently marked as eligible for inclusion, 68 articles had at least one taxon-species combination that both coders identified and could be aligned for inter-rater reliability (IRR) analysis after correcting minor spelling inconsistencies. Within these 68 articles, 107 taxon-species-group entries could be matched between coders for cell-level IRR analysis. Unmatched entries occurred when one coder extracted species-level or group-level information that the other did not capture. For matched entries, coders' responses on task design variables were compared. When multiple groups were tested for the same species within an article, entries were matched based on the order they appeared in the source material, as recorded by both coders.

IRR for 11 nominal variables—including cue use and overtraining indicators—was assessed using Gwet's AC1 via the *irrCAC* package (Gwet, 2014). For six numeric variables—including sample size, number of learning trials, and criterion percentage—unweighted Cohen's κ was computed using the *irr* package (Gamer et al., 2019), treating values as discrete codes where only exact matches were considered agreement. Agreement levels ranged from moderate to very good (see Supplementary Materials; S1 Table). For the variable concerning the number of trials available to reach the reversal criterion (i.e., reversal evaluation length)—the only one with moderate interrater agreement, and therefore flagged for manual resolution prior to use in any quantitative analyses—two lead coders reviewed the cell-level discrepancies between original and double-coded values (n = 35 cells). In cases where both lead coders agreed with either the original or the double-coded value (n = 33 cells), that value was retained. In the remaining two cases, where lead coders did not agree with either the original or double-coded entry, those cells were excluded from subsequent analyses, reducing the total number of cells from 271 to 269, and the total number of articles from 208 to 206. A detailed log of this process can be found in the shared code. All subsequent analyses used this reconciled dataset.

The final coding scheme and variable definitions are presented in Table 2, with additional details provided in S2 Table (Supplementary Materials). The complete dataset, analysis script, supplementary materials, and figures are available on Figshare: [a public link will be made available upon acceptance].

Data Analysis Approach

All data processing and statistical inference were carried out in R 4.5.1 (R Core Team, 2025) in RStudio (v2025.05.0+496) (Posit Team, 2025), using an R Markdown workflow to ensure reproducibility. Raw CSV files containing original and double-coded data were imported and type-harmonized using functions from the *tidyverse* suite (Wickham et al., 2019).

Descriptive statistics were computed by taxonomic group. Due to non-normal distributions, group differences were tested using the Kruskal–Wallis rank-sum test. Where significant, pairwise comparisons were performed using Dunn's test from *FSA* (Ogle et al., 2025), with *p*-values adjusted via the Benjamini–Hochberg procedure. For categorical design variables, global differences were tested via Pearson's χ^2 test on $k \times 2$ tables, using Monte Carlo simulation (10,000 replicates) when assumptions were violated. Post-hoc pairwise testing was conducted only for the binary variable indicating whether overtraining was used during learning, using Fisher's exact test with correction.

To explore multivariate structure, we performed two Linear Discriminant Analyses (LDA) with *MASS* (Venables & Ripley, 2002). In the first, six binarized cue variables (size, shape/pattern, color/brightness, object, spatial, olfactory) predicted taxa; in the second, three binarized dependent variable indicators (learning speed, overall performance, response speed) served as predictors. Discriminant functions were interpreted using loadings and visualized via scaled arrows. The explained variance for LD1 and LD2 was extracted to label axes, and ellipses (95% multivariate normal) and centroids were plotted for visual inspection.

All primary analyses were complemented by robustness checks that account for non-independence due to article-level clustering. For cell counts by taxon, we fitted a Bayesian multinomial model with article-level random effects. For percentage outcomes, we applied one-inflated beta regression models with article-level random intercepts. For binary outcomes, we used mixed-effects logistic regression. For numeric variables not normally distributed (e.g., sample size and number of learning trials), we ranked values and fitted linear mixed-effects models with article-level random intercepts. Robustness results are reported in each Results subsection and complete details can be found in our Figshare repository. LDA analyses were treated as exploratory visualizations and did not include clustering corrections.

Taxonomic Grouping Strategy

To characterize the diversity of species represented in reversal learning studies, we initially coded each study using a fine-grained taxonomy comprising 18 animal taxa, presented here alphabetically by common and scientific name: amphibians (Amphibia), arachnids (Arachnida), birds (Aves), canids (Canidae), cephalopods (Cephalopoda), cetaceans (Cetacea), euplerids (Eupleridae), felids (Felidae), fishes (Actinopterygii, Chondrichthyes, Coelacanthimorpha, and Cyclostomata), humans (*Homo sapiens*), insects (Insecta), musteloids (Musteloidea), non-human primates (Primata excluding *H. sapiens*), pinnipeds (Pinnipedia), reptiles (Squamata), rodents (Rodentia), ungulates (Ancodonta, Perissodactyla, Ruminantia, Suina, and Tylopoda), and ursids (Ursidae). This categorization aimed to maximize taxonomic specificity from the outset. Three taxa—cetaceans, felids, and ursids—were not represented in any retained articles, resulting in 15 taxa for further analysis.

To facilitate cross-group comparisons while preserving biological coherence, the 15 remaining taxa were aggregated into six broader groups based on sample size considerations and taxonomic associations: invertebrates (i.e., arachnids, cephalopods, and insects), fishes, amphibians, reptiles, birds, and mammals (i.e., canids, euplerids, humans, musteloids, non-human primates, pinnipeds, rodents, and ungulates).

Subsequent inspection revealed a pronounced overrepresentation of mammalian cells, which accounted for 147 of the 269 entries (54.65%). Within this group, three taxa—humans (n = 51), rodents

(n = 45), and non-human primates (n = 27)—accounted for approximately 83.67% of all mammalian cells. To reflect this internal structure while preserving interpretability, mammals were subdivided into four categories: rodents, non-human primates, humans, and other mammals (canids, euplerids, musteloids, pinnipeds, and ungulates).

Beyond mammals, one additional adjustment was made to address data sparsity. Amphibians were represented by just three cells across two articles, limiting the interpretability of this group on its own. To ensure sufficient sample size while maintaining biological relevance, amphibians and reptiles were merged into a single category based on shared physiological and phylogenetic characteristics (Meyer & Zardoya, 2003; Pough, 1983).

The final classification—referred to as *taxon_revised* in our dataset—comprised eight high-level groupings: invertebrates, fishes, amphibians and reptiles, birds, rodents, non-human primates, humans, and other mammals. Importantly, this final classification was not determined by formal taxonomic rank (e.g., class or order) but emerged from a bottom-up, data-driven process that balanced taxonomic coherence with the need for meaningful and interpretable statistical comparisons. Given the uneven representation of taxa and the descriptive nature of our review, we prioritized empirical coverage and analytical utility over strict phylogenetic hierarchy, ensuring sufficient within-group sample sizes for valid cross-taxon analysis.

Results and Topic Discussions

Taxonomic Representation and Sampling Disparities

Which Taxa are Most Frequently Studied?

To characterize how research effort has been distributed across taxa, we examined the number of studies per group using our final taxonomic classification (Figure 3). The analysis showed a pronounced imbalance, $\chi^2(7) = 31.40$, $p < .001$. Humans, alongside birds and rodents

($n = 51$, $z = 3$; $n = 47$, $z = 2.31$; and $n = 45$, $z = 1.96$ respectively) dominated the literature, even after mammals were subdivided into four distinct groups to mitigate overrepresentation. These taxa showed standardized residuals near or above 2, indicating significant or marginal overrepresentation relative to a uniform distribution. In contrast, invertebrates ($n = 19$, $z = -2.52$), amphibians and reptiles ($n = 24$, $z = -1.66$), and other mammals ($n = 24$, $z = -1.66$) were underrepresented.

To assess the robustness of these results and account for article-level clustering, we fit a Bayesian multinomial (categorical) model with taxon identity as the outcome and DOI (Digital Object Identifier) as a random effect. The model estimated the probability of each cell being assigned to a given taxon across all eligible observations. The posterior probability for humans (0.18), birds (0.18), and rodents (0.17) exceeded that of other taxa. Fishes (0.12) were represented approximately as would be expected for a uniform distribution, while non-human primates (0.10), amphibians and reptiles (0.09), other mammals (0.09) and invertebrates (0.07) all fell well below the uniform benchmark (0.125) that would indicate equal representation across taxa. Model diagnostics indicated good convergence and satisfactory fit based on posterior predictive checks.

Taken together, both the chi-squared analysis and the multinomial model provide converging evidence for pronounced and persistent taxonomic bias in the literature. Even after restructuring our classification to reduce mammalian overrepresentation and promote balance, and after accounting for clustering within studies, fundamental disparities in research attention persist. The literature remains disproportionately focused on a narrow set of taxa aligned with traditional model systems in comparative cognition.

How Well Are Species Represented Within Each Taxon?

When we looked at species-level representation, the disparities became more striking. To estimate taxonomic coverage, we used species counts from the NCBI Taxonomy Browser (https://www.ncbi.nlm.nih.gov/taxonomy; accessed May 14, 2025), a curated molecular database (Federhen, 2012). For each of the broad taxonomic groups included in this review, we calculated the proportion of known species represented in reversal learning studies relative to the total number of species in the relevant clades included within that grouping. Beyond humans (with one single species accounting for 100% of the data), non-human primates had the highest non-trivial coverage at 2.86% (14 of 490 known species). Other mammals (canids, euplerids, musteloids, pinnipeds, and ungulates) followed at 2.17% (11 of 506), then rodents at 0.39% (9 of 2,340). Birds (24 of 10,611) and amphibians and reptiles (21 of 16,130) each fell below 0.25%, and fishes were represented by only 0.05% of known species (13 of 24,311). For groups of invertebrates where we had data (arachnids, cephalopods, and insects)—these were represented by just 7 of over 172,000 species (0.004%), despite comprising the vast majority of described animal life. These results reveal that reversal learning research not only focuses on a narrow set of taxonomic groups, but within those groups relies heavily on repeated sampling of a small subset of species, severely limiting the taxonomic breadth available for comparative inference (see Supplementary Materials, S3 Table).

How Large are the Samples Used to Study Each Taxon?

To assess the sample scale, we compared sample size at two levels: per test group (cell size) and article (study-level sample size, restricted to single-taxon studies). Significant variation emerged in cell size across taxa, $\chi^2(7) = 37.41$, $p < .001$ (see Figure 4). A robustness check using a mixed-effects model with a random intercept for article (DOI) confirmed significant taxon-level disparities in cell size (Type III Wald $\chi^2(7) = 21.91$, $p = .003$), with all taxa except invertebrates and fishes tested at significantly smaller scales than humans. The study-level sample size comparison was non-significant ($\chi^2(7) = 12.84$, $p = .08$).

Humans were tested not only frequently but also at large scales: their mean cell size was 69.0, with a median of 46.0 ($n = 50$ cells), and their median study-level sample size was 47.0 ($n = 45$ studies). These values were closely aligned, suggesting relatively consistent sample sizes across cells and studies. Birds showed similarly high mean cell sizes ($M = 60.2$, $n = 46$ cells), but their median was far lower ($Mdn = 20.5$), indicating a right-skewed distribution. Their study-level median sample size ($Mdn = 26.0$, $n = 39$ studies) was higher than their median cell size, suggesting that some studies included multiple smaller groups or a few large ones. Invertebrates had the highest mean cell size overall ($M = 116.5$, $n = 19$), but this was driven by a small number of outliers. Their median cell size ($Mdn = 27.0$) and median study sample ($Mdn = 47.0$, $n = 11$) paint a more moderate picture of sampling effort in this group.

Other taxa were typically tested at smaller scales. Non-human primates had a mean cell size of 21.3 and a median of 18.0 ($n = 27$), with a median study sample of 24.0 ($n = 16$), suggesting modest and relatively consistent designs. Rodents ($Mdn = 18.0$, $n = 43$ cells) and other mammals ($Mdn = 16.0$, $n = 24$) showed similar patterns in cell size. However, rodents had a substantially higher median study sample ($Mdn = 36.0$, $n = 33$) than their cell-level median, resulting in a larger absolute difference ($\Delta = 18.0$). Amphibians and reptiles had the lowest overall cell-level sampling ($Mdn = 17.5$, $n = 24$), with a median study sample of 25.5 ($n = 10$). Fishes presented an intermediate case. Their mean cell size was 44.9 ($Mdn = 32.5$, $n = 32$), and their median study sample was 48.0 (n = 27), indicating that this group was typically tested with larger samples than other underrepresented taxa. Notably, their median cell and study sizes were almost equivalent, suggesting relatively stable within-study design choices.

The absolute difference between median cell size and median study sample size (Δ) varied widely across groups, from 1.0 in humans to 20.0 in invertebrates. Larger discrepancies may reflect either within-study replication (i.e., multiple small cells) or the presence of outlier designs with unusually large or small groups. These nuances are critical, as they shape both statistical power and the interpretability of group-level comparisons.

Do Studies Compare Taxa Directly, or Are They Siloed by Species?

To assess whether reversal learning research employs direct cross-taxa comparisons or remains confined within taxonomic boundaries, we examined the taxonomic scope of individual studies. The structure of the literature reveals extreme taxonomic siloing: 99% of the included studies (204 of 206) focused on a single major taxon from our taxon_revised classification, with only two studies adopting a multi-taxa design. Among the single-taxon studies, four lacked sample size data entirely. This overwhelming dominance of within-taxon research—combined with the near-absence of direct cross-taxa comparisons—compounds the asymmetries we observed in both research effort and species representation. The fragmented approach limits our ability to draw meaningful conclusions about cognitive similarities and differences across taxa, as most inferences about cross-taxa patterns rely on indirect comparisons between studies that used different methodologies, protocols, and analytical approaches.

Learning Phase Design Across Taxa

We examined how learning success was operationalized across taxa by analyzing two key design parameters: (1) the *length* of the evaluation window, defined as the number of consecutive trials over which performance was assessed to determine criterion attainment, and (2) the *accuracy* threshold, or the percentage of correct responses required within that window (see Table 4). Importantly, the evaluation window does not reflect the total number of trials administered during the learning phase, but rather the segment used to judge whether learning has occurred. This window may have been repeated until success was achieved or embedded within a longer training sequence.

How Large Are the Evaluation Windows for Learning Success Across Taxa?

We first tested whether the length of the evaluation window varied systematically across taxa. There were significant differences across groups, $\chi^2(7) = 37.42$, $p < .001$. The longest evaluation windows were found in birds and humans, each with a median of 20.0. Rodents (*Mdn* = 18.0), non-human primates (*Mdn* = 12.0), other mammals (*Mdn* = 10.0), fishes (*Mdn* = 10.0), and invertebrates (*Mdn* = 10.0) were typically evaluated over shorter spans. The shortest windows were used in amphibians and reptiles (*Mdn* = 6.0).

Post hoc comparisons revealed pronounced disparities in the length of evaluation windows across taxa. Amphibians and reptiles were consistently assessed with significantly shorter windows than birds ($p < .001$), rodents ($p < .001$), humans ($p < .001$), non-human primates ($p = .006$), other mammals ($p = .034$), and fishes ($p = .042$). In addition, invertebrate studies used shorter windows than studies on birds ($p = .007$), rodents ($p = .005$), and humans ($p = .019$). These patterns confirm that the behavioral evidence required to declare learning success was not evenly applied across taxa. Some groups, particularly amphibians and reptiles, and invertebrates, were subject to substantially more compressed evaluation windows than others, raising concerns about the comparability of success criteria in reversal learning tasks.

A robustness check using a mixed-effects model with a random intercept for article (DOI) found no evidence of taxon-level disparities in evaluation window length (Type III Wald $\chi^2(7) = 12.60$, $p = .08$). These findings suggest that cross-taxon variation in evaluation window length is not robust once large variation in window length between studies is taken into account.

How Strict are the Accuracy Thresholds Used to Define Learning Success?

We tested whether the percentage of correct responses required to reach the learning criterion differed across taxa. Although the global nonparametric test did not detect significant group differences, $\chi^2(7) = 11.22$, $p = .129$, a model-based robustness analysis using a one-inflated beta regression with article-level random intercepts revealed several taxon-level effects with strong evidence of deviation from the human reference. Specifically, other mammals ($\beta = -0.67$, 95% CI [−1.05, −0.29]), invertebrates ($\beta = -0.55$, 95% CI [−1.10, −0.01]), fishes ($\beta = -0.41$, 95% CI [−0.77, −0.05]), and rodents ($\beta = -0.39$, 95% CI [−0.64, −0.12]) were consistently evaluated using lower learning criteria than humans, with 95% credible intervals excluding zero. In contrast, estimates for birds, amphibians and reptiles, and non-human primates included zero, indicating limited or inconclusive evidence for taxon-level differences compared to humans. Posterior predictions by taxon, including model-based means and medians, are available in our open repository alongside all data and code.

Descriptive statistics further indicated that median learning thresholds were clustered around 80–84% for most taxa (see Figure 5, Panel A), with humans held to a stricter standard

(*Mdn* = 90%) and amphibians and reptiles showing the tightest central tendency (*M* = 84.0%, *Mdn* = 83.3%). Notably, invertebrates (Δ = 6.7), fishes (Δ = 5.4), other mammals (Δ = 4.6), and birds (Δ = 3.9) exhibited the largest mean–median discrepancies, suggesting greater within-group heterogeneity and more variable criteria in these taxa. The overall median was 83%, underscoring a general benchmark, but persistent within-group variation highlights ongoing methodological inconsistencies.

Collectively, the results of this section reveal that learning success has not been operationalized uniformly across taxa, with design choices likely reflecting assumptions about cognitive endurance, task structure, or species-specific methodological traditions. Humans were consistently held to stricter standards, while several other taxa were assessed using more lenient accuracy thresholds and shorter evaluation window lengths. Within-group distributions were often asymmetrical. Such variation in how learning is defined—whether through strict sustained performance or lenient, brief windows—has direct implications for comparability.

Is Overtraining Used Consistently Across Taxa?

To examine whether overtraining practices differ systematically across taxa, we coded the presence or absence of overtraining procedures in each study cell. The observed frequencies reveal pronounced disparities: overtraining was implemented in 67% of amphibian and reptile cells, but was much less common in birds (30%), fishes (25%), and non-human primates (22%). Use of overtraining was rare for humans (12%), rodents (9%), other mammals (8%), and invertebrates (5%). These differences were statistically significant, $\chi^2 = 44.15$, $p < .001$, B = 10,000. Pairwise comparisons confirmed that amphibians and reptiles were significantly more likely to be overtrained than all other taxa (all adjusted *p*s < .05). No other contrasts reached significance after correction. As a robustness check, we fitted a generalized linear mixed-effects model (GLMM) for overtraining presence, with taxon as a fixed effect and study DOI as a random intercept. The model revealed no overall effect of taxon on the likelihood of overtraining procedures ($\chi^2(7) = 5.91$, $p = .550$). Most taxon-level contrasts were not significant, and confidence intervals were wide.

Taken together, these results suggest that overtraining procedures appear to be unevenly implemented across taxa at the descriptive level. However, when accounting for study-level clustering in a mixed-effects model, most taxon-level differences are no longer statistically significant, indicating that much of the apparent taxon-level variation in overtraining procedures may be driven by methodological choices made at the study level, rather than by systematic differences in taxon-specific research practices. Nonetheless, these methodological choices themselves may reflect taxon-specific traditions or assumptions—such as differences in views about memory consolidation, behavioral variability, or experimental tractability—as well as shifts in research focus over time. For example, for some taxa, questions about the effects of overtraining might have been thoroughly examined in earlier decades (Hicks, 1964; Mackintosh, 1969), reducing the incentive to revisit them in recent work. Thus, while observed taxon-level disparities in overtraining are largely accounted for by between-study variation, these patterns still highlight how evolving design conventions shape what is measured and how cross-species results are interpreted.

Reversal Phase Design Across Taxa

How Many Reversal Phases Are Administered Across Taxa?

To assess whether the complexity of reversal learning protocols varies systematically across taxa, we examined the number of reversal phases administered in each study. Studies differed substantially, with reversal counts ranging from a single phase to 176, the latter representing an extreme case. Across taxa, most studies used only one reversal, and the prevalence of designs with multiple reversals varied substantially across groups. Single-reversal designs were overwhelmingly common in amphibians and reptiles (87.5%), fishes (83.9%), and invertebrates (78.9%). In contrast, only about half of non-human primate studies (53.8%) and human studies (51.1%) employed single reversal designs. A statistical comparison confirmed significant differences across taxa, $\chi^2(7) = 20.53$, $p = .005$. Post hoc comparisons showed that humans underwent significantly more reversals than fishes ($p = .028$) and amphibians and reptiles ($p = .042$). No other pairwise contrast remained significant after correction, although several showed marginal differences. A robustness check using a linear mixed-effects model that included study DOI as a random intercept did not find significant taxon-level variation in the number of reversals administered (Type III Wald $\chi^2(7) = 13.03$, $p = .071$). Model coefficients echoed the main finding of fishes receiving significantly fewer reversal phases than humans, but also found differences between humans and birds after accounting for potential clustering by study.

These findings suggest that mammalian taxa—particularly non-human primates and humans—are more often studied using interreversal designs, potentially enabling assessment of learning-set formation and behavioral stability. In contrast, studies of amphibians and reptiles, fishes, and invertebrates typically use only a single reversal, which limits the ability to examine sustained behavioral flexibility or the development of meta-learning strategies. This methodological disparity may contribute to apparent cognitive differences between taxa, as multi-reversal protocols provide richer data on adaptive learning processes that single reversals cannot capture.

How Large Are the Evaluation Windows for Reversal Success Across Taxa?

To determine whether the stringency of reversal learning assessment varies across taxa, we examined the number of trials over which reversal success was evaluated. Evaluation window lengths during the reversal phase tended to be short overall and varied meaningfully between groups, $\chi^2(7) = 36.58$, $p < .001$. In 41 out of 226 cells (18.1%), the number of trials used to evaluate reversal was not the same as for learning—directly demonstrating that nearly one in five studies applied different standards across phases. Within-group correlations provided further nuance regarding the degree of correspondence between evaluation window lengths during learning and reversal phases. For all taxa except fishes, Pearson correlations ranged from $r = 0.649$ to $r = 0.998$ (all *p*s < .001). Fishes, however, showed a weaker, non-significant correlation ($r = 0.326$, $p = .07$). Inspection of the data for this group revealed cases where reversal phase length substantially exceeded learning phase length, suggesting potential outlier influence. Winsorized correlations confirmed that after accounting for extreme values, the correlation for fishes ($r = 0.691$, $p < .001$) aligned with the pattern observed in other groups. Taken together, these results indicate that all groups demonstrated moderate to strong correspondence between learning and reversal phase window lengths, though the strength of this association varied across groups, pointing to meaningful variation in methodological implementation across taxa.

The shortest windows were again found in amphibians and reptiles (*Mdn* = 6.0), which were significantly shorter than those used for birds (*Mdn* = 20.0), humans (*Mdn* = 20.0), rodents (*Mdn* = 15.0), non-human primates (*Mdn* = 12.0), and fishes (*Mdn* = 11.0) (all adjusted *p*s < .01). Invertebrates (*Mdn* = 10.0) also had relatively short windows, significantly shorter than those used in humans ($p = .043$), and marginally shorter than those used in birds ($p = .051$).

A robustness check using a linear mixed-effects model, which included a random intercept for study DOI, found little evidence of taxon-level variation in evaluation window length (Type III Wald $\chi^2(7) = 12.99$, $p = .07$). The model did however find that amphibians and reptiles—and marginally invertebrates and other mammals—were evaluated with significantly shorter windows than humans, demonstrating that some of the differences found in the non-parametric analysis are robust to potential clustering within studies.

These findings reveal that reversal learning success is assessed using markedly different standards across taxa. Groups with shorter evaluation windows require fewer consecutive correct responses to demonstrate reversal mastery, potentially creating an illusion of superior flexibility. Conversely, taxa evaluated over longer windows must sustain accurate performance across more trials, providing stronger evidence of genuine learning but potentially making reversal appear more difficult. This methodological inconsistency means that apparent differences in reversal learning speed or success between taxa may reflect differences in assessment stringency rather than true cognitive variation, as also suggested by the varying degrees of correspondence between learning and reversal window lengths across taxa.

How Strict are the Accuracy Thresholds Used to Define Reversal Success?

To assess whether the stringency of performance criteria varies across taxa, we examined the percentage-correct thresholds used to define successful reversal learning. Percentage-correct thresholds for reversal success were largely aligned across taxa (Figure 5, Panel B). In only 8 out of 213 cells (3.8%), the performance threshold used for reversal differed from the one used for learning. As with learning criteria, most groups clustered around a median of 80–84%, pointing to a shared operational benchmark. Other mammals showed the highest median threshold (88.0%), followed by amphibians and reptiles (83.3%), humans (83.2%), non-human primates (83.0%), and rodents (81.7%). Invertebrates, fishes, and birds had the lowest (80.0%). The overall median across taxa was 83.0%. Some groups, particularly invertebrates ($\Delta = 8.0\%$) and other mammals ($\Delta = 6.3\%$), showed asymmetries between mean and median values, pointing to heterogeneity in criteria across studies.

The omnibus non-parametric test found no significant group differences ($\chi^2(7) = 6.60$, $p = .471$). As a robustness check, we fitted a one-inflated beta mixed-effects model with article-level random intercepts. This model found that other mammals ($\beta = -0.46$, 95% CI [−0.91, −0.02]) were held to credibly lower reversal thresholds relative to humans. For all other comparisons to humans, the 95% credible intervals included zero, indicating no clear evidence for taxon-level differences after accounting for study-level variation (see open repository for posterior predictions).

Taken together with our previous findings, a nuanced picture emerges: while researchers generally agree on what constitutes adequate performance (setting median thresholds between 80-88% correct trials across taxa), they differ substantially in how that performance is assessed.

Taxa vary markedly in the number of reversals administered and the length of evaluation windows—structural factors that fundamentally shape how behavioral flexibility is demonstrated and measured. Single-reversal tasks provide only a narrow snapshot of behavioral updating, whereas multi-reversal designs can reveal patterns of adaptation, learning-set formation, and meta-cognitive development.

Cross-Taxon Patterns in Cue Use and Outcome Measurement

Are Different Taxa Studied Using Different Types of Cues?

To explore whether cue configurations differed systematically across taxonomic groups, we conducted a Linear Discriminant Analysis (LDA) using six binarized cue variables: size, shape/pattern, color/brightness, object identity, spatial location, and olfactory cues. While LDA assumes multivariate normality and homogeneity of covariance—assumptions not fully met by binary predictors—it remains useful for visualizing group separation and cue structure. Our goal was descriptive, namely to characterize how cue use varies across taxa. The LDA model achieved an accuracy of 39.4% (95% CI [33.5%, 45.5%]), which was significantly higher than the no-information rate of 19% ($p < .001$). This demonstrates that the predictor variables successfully discriminate among the eight classes.

The first two discriminant functions accounted for 76.4% of between-group variance (LD1 = 42.4%, LD2 = 34%). LD1 captured a primary contrast between visual-structural cues and olfactory cues. Specifically, LD1 loadings were strongly negative for size (−2.93), shape/pattern (−2.09), and object identity (−1.62), and strongly positive for olfactory cues (+3.12), indicating that taxa positioned lower on LD1 were more frequently studied using visual discrimination tasks, whereas higher LD1 values reflected olfactory-based paradigms. Spatial cues contributed modestly to LD1 (+0.32). LD2 distinguished color/brightness cue use from olfactory and spatial modalities. Color/brightness cues loaded positively on LD2 (+1.24), while spatial (−1.65) and olfactory (−1.89) cues loaded negatively, suggesting this axis separated taxa that rely on hue-based discrimination from those tested in spatial or chemosensory contexts (see Table 5).

The LDA results revealed clear taxonomic separation based on the types of information animals were trained to use. In the LDA space (Figure 6), fishes and birds clustered in the upper-right quadrant, closely aligned with color/brightness cues (high LD2). Invertebrates and rodents occupied the lower-right quadrant, associated more with olfactory and spatial cues (positive LD1, negative LD2). Amphibians and reptiles fell similarly in this region, though with a tighter distribution. In contrast, non-human primates and humans were positioned left of center on LD1, reflecting stronger use of visual-structural cues such as shape, size, and object identity. Other mammals were distributed near the center, suggesting more heterogeneous cue use or less reliance on any single modality.

These findings indicate that cue selection in reversal learning tasks systematically varies by taxon, reflecting both species-specific sensory capabilities and researchers' assumptions about what animals can perceive and process. Design choices represent more than technical adaptations—they embody assumptions about what sensory modalities are most relevant or tractable for particular animal groups. These assumptions have important implications: when cue modalities align with a species' dominant sensory systems, learning may appear more efficient—not necessarily because of greater behavioral flexibility, but because the perceptual demands are more easily met. Consequently, apparent species differences in cognitive performance may conflate genuine cognitive variation with cue compatibility. To address this confound, future studies should either standardize cue types across taxa or systematically manipulate cue modalities to isolate their effects on performance, enabling more valid assessments of behavioral flexibility across species.

Are Certain Outcome Measures Reported More Often for Specific Taxa?

We conducted a second LDA to explore whether the types of dependent variables used in reversal learning studies vary systematically across taxa. The model included three binary-coded dependent variables: learning speed, overall performance, and response speed. These were converted into a numeric format for analysis. Model accuracy reached 27.5% (95% CI [22.3%, 33.3%]), significantly exceeding the no-information rate of 19% ($p < .001$). The reduced performance relative to the first model (39.4%) suggests less clear differentiation among groups with these predictors, indicating more overlapping group profiles.

The first two discriminant functions accounted for 88.5% of the between-group variance (LD1 = 66.5%, LD2 = 22.0%), capturing substantial variation in reporting conventions. LD1 was driven primarily by use of learning speed as a dependent variable, which had a strong positive loading (+2.27). Taxa positioned further along the LD1 axis—such as fishes and amphibians & reptiles—were more frequently associated with studies reporting learning speed. In contrast, invertebrates and humans scored lower on LD1, indicating that learning speed was less commonly reported in those groups (see Table 5).

LD2 distinguished taxa based on their use of overall performance (+2.00) and response speed (+1.20). Higher LD2 scores reflect stronger emphasis on these outcome measures. Interestingly, invertebrates, as well as amphibians and reptiles, scored relatively high on LD2. In contrast, birds and non-human primates scored lower—suggesting that, contrary to expectations, performance and latency measures were more commonly reported in studies of non-model taxa than in those involving traditionally dominant model groups like birds and non-human primates.

In the discriminant space, fishes and amphibians and reptiles clustered in the upper-right quadrant, reflecting frequent use of both learning speed and overall performance as dependent variables. Invertebrates and humans appeared in the upper-left, more associated with overall performance and response speed, but not learning speed. Birds and non-human primates were positioned near the center or lower quadrants of the discriminant space, indicating they are not strongly associated with any particular outcome measure. This suggests a more uniform or limited pattern of reporting, with fewer studies using diverse metrics like learning speed, latency, or performance composites. Rodents and other mammals occupy overlapping mid-range positions, suggesting mixed or less standardized dependent variable use.

These patterns reveal marked variation in how outcomes are measured—and that these differences track with taxonomic lines. Without accounting for such variation, cross-species comparisons risk conflating cognitive performance with methodological conventions. Greater alignment or transparency in the selection of dependent variables may be necessary to ensure valid comparisons of behavioral flexibility across taxa.

General Discussion

Our systematic review of 206 reversal learning studies revealed three fundamental limitations that compromise the reliability and comparability of current findings. First, research effort is severely imbalanced. Second, research remains taxonomically siloed, with 99% of studies testing only one taxonomic group per study. Third, methodological standards vary dramatically across taxa, with some groups requiring three times more evidence than others to demonstrate "successful" learning—a disparity that likely extends beyond reversal learning and may reflect broader patterns in how cognitive performance is operationalized across comparative cognition research. Together, these limitations force researchers to rely on indirect comparisons across methodologically inconsistent studies to draw conclusions about cognitive differences between taxa.

The Magnitude of Methodological Bias

What does it mean to compare behavioral flexibility across species when the tools we use to measure it vary so widely? In reversal learning—a canonical task in comparative cognition—shared terminology often conceals profound differences in experimental design. From sampling and cue selection to performance thresholds and outcome reporting, protocols diverge in ways that fundamentally reshape what is being asked of different species.

At first glance, reversal learning appears to offer a common currency for assessing behavioral flexibility. On closer inspection, however, it becomes clear that this currency is not uniformly minted across taxa. Reversal learning research is dominated by a few groups—humans, birds, and rodents—with well-established experimental traditions. Together, these three taxonomic groups represent 53.16% of all cells in our dataset (143 out of 269), yet they encompass only 34 individual species. These groups are typically tested using visually rich stimuli, extended evaluation windows, and multiple reversals. Other taxonomic groups—particularly invertebrates, fishes, and amphibians and reptiles—are studied under leaner designs, with smaller samples, shorter evaluation windows, fewer reversals, and cue types tailored to presumed sensory modalities. These discrepancies are not peripheral; they reflect underlying assumptions about what counts as learning, how behavior stabilizes, and which dependent variables merit attention. Although we observed a broad convergence around an 80% accuracy threshold for defining successful learning and reversal performance, this consistency stands in contrast to the striking variability in other core aspects of study design.

The scale biases are striking. For example, amphibians and reptiles must demonstrate reversal success over just six trials on average, while humans and birds are evaluated over 20 trials. Similarly, overtraining is applied to 67% of amphibian and reptile studies but fewer than 12% of human studies, fundamentally altering the learning context. Even species representation within taxa reveals dramatic inequalities: while humans represent 100% of their taxonomic category by definition, invertebrates—despite comprising over 95% of described animal species—are represented in reversal learning research by just 0.004% of known species within the invertebrate clades included in our dataset.

Such variation is not merely technical—it is interpretive, and it introduces systematic measurement bias. For instance, the psychometric properties of “successful reversal learning” differ fundamentally depending on design parameters. A correct response after 6 trials at an 80% threshold requires 5 correct choices out of 6; the same criterion after 20 trials at 90% requires 18 out of 20. These are not equivalent measures of the same construct. The first permits substantially more error and requires far less sustained accuracy, making it both easier to achieve and more susceptible to chance performance. This echoes a longstanding concern in comparative cognition: that apparent species differences may reflect confounded variables—sensory, motor, motivational, or, as our data suggest, methodological—rather than true cognitive variation (Bitterman, 1965).

Methodological variation per se is not problematic—some adaptation to species-specific constraints is necessary and appropriate. The critical issue is that this variation tracks systematically with taxonomic boundaries, and because 99% of studies test only a single taxon, these differences are never directly evaluated. Different design choices do not necessarily measure entirely different constructs, but they do differentially weight various aspects of behavioral flexibility. Single-reversal designs assess whether an animal can update a learned association; multi-reversal designs additionally capture learning-set formation and meta-learning. Short evaluation windows may detect rapid behavioral updating but provide weaker evidence of stable, consolidated learning; longer windows provide stronger evidence of robust flexibility but may underestimate performance in species with faster learning dynamics or higher trial-by-trial variability. When these design asymmetries systematically align with taxonomy—and are never tested against alternative parameterizations within comparative frameworks—apparent species differences in behavioral flexibility may conflate true cognitive variation with differences in measurement strategies.

This reveals a core challenge: behavioral flexibility is not consistently operationalized, but instead reflects taxon-specific conventions and methodological choices. As a result, prevailing claims about cognitive hierarchies may be driven more by methodological asymmetries than by true biological differences. A deeper understanding of behavioral flexibility will remain elusive until these disparities are addressed and the field is supported by a more taxonomically inclusive and methodologically coherent data infrastructure. Despite decades of reflection on the limits of taxonomic bias, our findings suggest that the concerns raised by Beach 75 years ago (Beach, 1950)—particularly regarding the concentration of research on a limited number of species and behavioral patterns—remain relevant today.

The systematic differences we document naturally raise the question: why do these methodological asymmetries exist? There are several plausible explanations. Methodological variation may reflect genuine ecological and physiological constraints: species differ in sensory capabilities, attention spans, and the timescales over which they naturally learn, making certain protocols more appropriate for some taxa than others. Historical research traditions also play a role—established model organisms inherit protocols from decades of prior work, while newly studied groups often adopt methods from related species without systematic justification. Practical and animal welfare considerations such as housing constraints, resource availability, and regulatory frameworks further shape what is feasible. However, distinguishing among these possibilities—determining which differences represent necessary adaptations versus arbitrary conventions—requires empirical investigation beyond the scope of a systematic review. Such work would need to experimentally manipulate task parameters within and across species under controlled conditions. Our contribution is to establish that these asymmetries exist, are systematic, and must be addressed before cross-species cognitive comparisons can be interpreted with confidence.

A Roadmap for Reform

Acknowledging the complexity of cross-species comparison is an important step toward more rigorous and inclusive research. Moving forward, the challenge lies in rethinking how such comparisons are conducted in practice. Several influential studies have adopted large-scale, coordinated designs to tackle specific comparative questions—for instance, investigations of problem-solving in mammalian carnivores (Benson-Amram et al., 2016) and self-control in mammals and birds (MacLean et al., 2014). More recently, other large-scale collaborations, united under the common term “Big Team Science” (BTS) have emerged as a promising model in response to growing concerns about replicability and fragmentation in the behavioral sciences (Alessandroni, et al., 2024, 2025; Barrett et al., 2025; Battini et al. 2024; Baumgartner et al., 2023; Coles et al., 2022; Forscher et al., 2022; Prétôt et al., 2026; Vaidis et al., 2025). Examples of BTS collaborations that have yielded single-taxon empirical results are ManyBabies (e.g., ManyBabies Consortium, 2020, on infant-directed speech), ManyPrimates (e.g., ManyPrimates et al., 2022, on short-term memory in primates), ManyBirds (e.g., ManyBirds Project et al., 2025, on neophobia in birds), and ManyDogs (e.g., ManyDogs Project et al., 2023, on pointing comprehension in dogs).

These and other global networks unite hundreds of researchers across institutions and disciplines to co-design, implement, and analyze shared studies. By harmonizing methods across settings, they yield larger, more diverse datasets, increase statistical power, and improve reproducibility. For comparative cognition, they offer something more: a unique path toward coordinated, taxonomically inclusive comparisons that would be difficult—if not impossible—for individual teams to achieve alone.

ManyManys exemplifies this potential (https://manymanys.github.io/) (Alessandroni, Altschul, Baumgartner, et al., 2024; Alessandroni, Altschul, Bazhydai, et al., 2024). This cross-species/taxon BTS collaboration draws researchers from biology, psychology, zoology, philosophy, and allied fields, working with a broad array of species, to co-develop protocols that are both ecologically grounded and analytically rigorous. Crucially, the aim is not to eliminate variability altogether, but to shift from unstructured, legacy-driven differences to intentional, *species-fair* design. Rather than imposing strict uniformity, ManyManys promotes standardized frameworks that are flexible enough to respect taxon-specific constraints while still enabling meaningful cross-species comparisons (Farrar et al., 2021). In doing so, it directly addresses the design asymmetries mapped in this review—not by suppressing diversity, but by coordinating it. Importantly, BTS efforts do not seek to replace the rich tradition of species-focused research but to complement it by enabling coordinated comparisons that would be difficult to achieve otherwise. While such efforts may help address taxonomic imbalance, they do so only in part; broader structural changes remain essential to ensure more inclusive and representative cross-species research.

Immediate Actions

While large-scale collaborations are reshaping the field, individual researchers can take immediate steps to promote methodological equity. The field would likely benefit from greater standardization of evaluation criteria across taxa, potentially through the use of a consistent trial window as a baseline for comparison, with deviations explicitly justified based on species-specific constraints, not convenience. Whenever possible, various core outcome measures—such as learning speed, response speed, and overall performance—should be included to support comprehensive, unbiased cross-taxa comparisons and reduce the risk of selective reporting.

Researchers should explain why particular modalities were used, how they may influence performance, and whether alternatives were considered. This is especially important in cross-species work, where perceptual and ecological mismatches can obscure interpretation. Adopting protocols from well-studied species without careful adaptation risks undermining, rather than improving, comparability—particularly for understudied taxa. Extensive justification should be provided not only for cue types, but also for parameters such as timing, inter-trial intervals, and repetition schedules. To reduce flexibility in data analysis and enhance reproducibility, preregistration of protocols and analysis plans is encouraged—especially in multi-species studies, where inconsistencies are most likely to arise.

These steps, while modest in isolation, would collectively represent a meaningful shift toward more equitable, transparent, and interpretable comparative cognition research. For underrepresented taxa in particular, no adjustment is too small if it helps ensure that cognitive performance is measured on fair and ecologically grounded terms.

Limitations and Future Directions

While comprehensive, our review has important limitations that point toward future research priorities. We focused on English-language publications from a single database, potentially missing important work from other linguistic and cultural traditions. Publication bias toward positive results may further skew our understanding of which taxa appear “cognitively flexible.” Because we were primarily interested in ecologically valid experimental conditions, we excluded clinical, pharmacological, and genetic modification research. We focused on organism-level behavioral measures, excluding research on the neural mechanisms underlying reversal learning—a field that has expanded considerably in recent years (see Izquierdo et al., 2017). This neuroscience work employs distinct methodological approaches and faces different comparability challenges than behavioral studies, warranting separate systematic examination.

Similarly, we only considered articles published after 2013 to keep the number of sources manageable, although this decision necessarily excluded a body of relevant work from earlier decades. For instance, following Harlow’s pioneering contribution on learning sets (Harlow, 1949), several seminal studies on reversal learning were conducted between 1950 and 1980 and published in outlets as reputable as *Science* (Moffett & Ettlinger, 1966; Wiener & Huppert, 1968), *Psychonomic Science* (Kendler & Kimm, 1964; McDaniel, 1969), *Journal of Comparative and Physiological Psychology* (Gollin, 1964; Mackintosh, 1969; Reid, 1958; Warren, 1960), *Journal of Experimental Psychology* (Birch et al., 1960; Erlebacher, 1963), *Journal of Genetic Psychology* (Warren & Warren, 1962), and *Animal Behaviour* (Capretta & Rea, 1967). The lack of a broader temporal window also limits our ability to trace long-term trends in methodological implementation.

In addition, substantial gaps remain beyond our review's scope. Phylogenetic patterns in reversal learning are often assumed but seldom tested systematically. Socio-ecological factors—such as diet breadth, habitat complexity, or social structure—may relate to performance, but current datasets do not always allow for a thorough exploration of these potential links (Lambert & Guillette, 2021; Mettke-Hofmann, 2014). Within-species variation is similarly understudied: age, sex, rearing conditions, social context, and personality likely influence learning dynamics, yet their roles are rarely isolated (Griffin et al. 2015; Jeanson & Weidenmüller, 2014; Laskowski et al., 2022; Pryce et al., 2004; Thornton & Lukas, 2012). Similarly, developmental trajectories are poorly understood: how reversal performance evolves across the lifespan, and how it relates to broader cognitive profiles or heritable traits, remains an open question. Task complexity, too, deserves closer scrutiny—not just as a methodological parameter, but as a potential driver of learning and flexibility in its own right.

Several methodological issues also call for greater empirical anchoring. Cue modality likely interacts with species-specific sensory systems in ways that affect both accuracy and learning speed, but this interaction is rarely quantified. Even baseline acquisition—the threshold for initiating a reversal point at which learning is deemed sufficient to introduce a reversal—is unevenly operationalized across studies. We still lack consensus on how many trials constitute a stable acquisition, or whether this varies predictably across taxa. Clarifying these parameters is essential for improving both internal validity and comparative scope.

The observed asymmetries in task design raise deeper questions about the interpretability of species differences in behavioral flexibility. When tasks are unevenly structured across taxa—with, for example, more reversal phases or longer evaluation windows in mammals than in invertebrates—performance differences may reflect experimental affordances more than underlying cognitive capacities, which risks reinforcing perceived cognitive hierarchies that may be artefactual. If certain taxa repeatedly face more forgiving or ecologically aligned testing conditions, they may appear more flexible simply because their task constraints are more accommodating. Conversely, taxa tested under sparse or truncated protocols may underperform despite possessing comparable flexibility. These methodological biases not only skew our understanding of cognitive evolution but also obscure meaningful within-group variation, such as interindividual differences in learning rates or strategy use.

In sum, the field has reached a point where expanding taxonomic breadth must be matched by increased methodological reflexivity. If we are to compare behavioral flexibility across species, we must first address the flexibility built into our own designs (Audet & Lefebvre, 2017). Just as BTS offers a roadmap for improving transparency, logistical scale, and rigor in behavioral research, it may also prove essential for addressing the structural asymmetries that currently constrain cross-species inference. This review charts those asymmetries—and calls for a more critical, transparent, and taxon-conscious approach to measuring and comparing behavioral flexibility across the animal kingdom.

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Tables

Table 1

Key Research Questions and Findings

Question	Finding
Taxonomic representation and sampling disparities	
Which taxa are most frequently studied?	Humans, birds, and rodents are the most frequently studied groups. Invertebrates, amphibians and reptiles, and other mammals are underrepresented in the literature.
How well are species represented within each taxon?	Excluding humans, fewer than 3% of species are represented within each taxon.
How large are the samples used to study each taxon?	Sample sizes vary by taxon and level (cell vs. study). Humans show consistently large median sample sizes across levels.
Are cross-taxa comparisons common, or does research remain siloed by species?	Reversal learning research is taxonomically siloed: 99% of studies focus on a single taxon.
Learning phase design across taxa	
How large are the evaluation windows for learning success across taxa?	Humans and birds are assessed over more trials (*Mdn* = 20) than rodents (*Mdn* = 18) and other taxa (*Mdn* ≤ 12).
How strict are the accuracy thresholds used to define learning success?	Humans show stricter thresholds (*Mdn* = 90%) than other taxa (*Mdn* = 80–86%).
Is overtraining used consistently across taxa?	No. 67% of study cells involving amphibians and reptiles show overtraining, compared to ≤ 30% of study cells for other taxa.
Reversal phase design across taxa	
How many reversal phases are administered across taxa?	Single-reversal designs are the most prevalent across taxa, ranging from ~50% in human and nonhuman primates to > 80% in amphibians and reptiles and fishes.
How large are the evaluation windows for reversal success across taxa?	Similar to the learning phase, humans and birds are assessed over more trials (*Mdn* = 20) than rodents (*Mdn* = 15) and other taxa (*Mdn* ≤ 12) in the reversal phase.
How strict are the accuracy thresholds used to define reversal success?	Accuracy thresholds are consistent across taxa (*Mdn* = 80–88%). Fewer than 4% of studies use different thresholds for learning and reversal.
Cross-taxon patterns in cue use and outcome measurement	
Do different taxa show preferential association with different types of cues?	Yes, with birds and fish more strongly associated with color and brightness cues, invertebrates and rodents with olfactory and spatial cues, and humans and non-human primates with shape, size, and object identity cues.
Do specific taxa show preferential association with certain outcome measures?	Yes, with fish and amphibians and reptiles loading most strongly onto overall performance and learning speed, and humans and invertebrates onto overall performance and response speed. Remaining taxa showed more distributed loadings across outcome measures.

Table 2

Coding Scheme and Variable Definitions

Variable	Definition	Values
Animal taxon	Taxonomic group of the species studied.	Amphibians; Arachnids; Birds; Canids; Cephalopods; Cetaceans; Euplerids; Felids; Fishes; Humans; Insects; Musteloids; Non-human primates; Pinnipeds; Reptiles; Rodents; Ungulates; Ursids
Species name	Latin binomial name of the species.	e.g., *Homo sapiens*
Sample size	Number of individuals tested at the start of the learning phase (excluding dropouts).	e.g., 50
Reinforcement type	Type of consequence following a choice; main or direct reinforcer reported.	Reward – Food/water; Reward – Audio/visual; Reward – Social; Reward – Material; Reward – Safety/refuge; Punishment; Reward + Punishment; Not applicable
Learning criterion (%)	Percentage of correct choices required to meet the learning criterion in the learning phase.	e.g., 90; Not applicable (if varying or not reported)
Total learning trials	Number of trials available to reach the learning criterion.	e.g., 10; Not applicable (if varying or not reported)
Overtraining	Whether the learning phase continued beyond the learning criterion.	Yes; No
Size cue(s)	Whether the association involves size cues (e.g., choosing a larger vs. smaller stimulus).	Yes; No
Shape or pattern cue(s)	Whether the association involves shape or pattern cues (e.g., square vs. triangle).	Yes; No
Color or brightness cue(s)	Whether the association involves color or brightness cues (e.g., red vs. blue).	Yes; No
Object cue(s)	Whether the association involves object identity (e.g., specific object vs. another).	Yes; No
Spatial or location or position cue(s)	Whether the association involves spatial cues such as location or side (e.g., left vs. right).	Yes; No
Olfactory cue(s)	Whether the association involves odor cues (e.g., different scents).	Yes; No
Total or maximum reversal phases	Total number of reversal phases administered. If variable across subjects, the maximum number achieved is reported.	e.g., 1; Not applicable (if varying or not reported)
Reversal criterion (%)	Percentage of correct choices required to reach the reversal learning criterion.	e.g., 90; Not applicable (if varying or not reported)
Total reversal trials	Number of trials available to reach the reversal criterion.	e.g., 10; Not applicable (if varying or not reported)

Learning speed	Whether the study reports how quickly individuals reach reversal criterion (e.g., number of sessions, trials, or errors). Typical examples are: (i) number of sessions until reaching criterion (e.g., in single-reversal tasks); (ii) (average) number of trials until reaching criterion in the reversal phase(s); (iii) (average) number of errors until reaching criterion in the reversal phase(s).	Yes; No
Overall reversal performance	Whether the study reports overall reversal performance (e.g., percent correct or errors across reversal trials). Common examples are: (i) number or percentage of correct choices across reversal(s); (ii) number or percentage of incorrect choices across reversal(s); and (iii) number or percentage of correct choices in the first XYZ reversal trials.	Yes; No
Speed of response or choice during reversal(s)	Whether the study reports how quickly individuals make a choice during the reversal(s), including latency (i.e., time from stimulus to response) and movement duration. Common examples are: (i) latency (i.e., time from stimulus presentation to response) and (ii) other measures of response speed, such as movement duration.	Yes; No

Table 3

Sample Size Metrics Across Taxa

Taxon	Sample size per cell			Sample size per study			
	M	*Mdn*	*n*	*M*	*Mdn*	*n*	Δ *Mdn* (study-cell)
Invertebrates	116.5	27.0	19	201.3	47.0	11	20.0
Fishes	44.9	32.5	32	53.3	48.0	27	15.5
Amphibians & reptiles	22.7	17.5	24	54.5	25.5	10	8.0
Birds	60.2	20.5	46	70.8	26.0	39	5.5
Rodents	30.0	18.0	43	38.5	36.0	33	18.0
Other mammals	35.2	16.0	24	44.0	22.0	19	6.0
Non-human primates	21.3	18.0	27	36.0	24.0	16	6.0
Humans	69.0	46.0	50	76.2	47.0	45	1.0

Note. Cell-level sample size refers to the average number of individuals per tested group; study-level sample size reflects the average number of individuals per study (for single-taxon studies). The Δ median shows how much these values differ within each taxon. Larger differences suggest that some taxa were more often studied using multiple small groups or uneven sampling. This can affect statistical power and increase variability in results, which in turn may limit the reliability of estimates and complicate comparisons across taxa.

Table 4

Summary of Learning and Reversal Task Parameters Across Taxa

Taxon	Learning phase				Reversal phase					
	Learning window length (N trials)		Learning success criteria (% trials)		Number of reversal phases		Reversal window length (N trials)		Reversal success criteria (% trials)	
	M	*Mdn*	*M*	*Mdn*	*M*	*Mdn*	*M*	*Mdn*	*M*	*Mdn*
Invertebrates	12.2	10.0	73.3	80.0	2.3	1.0	12.0	10.0	72.0	80.0
Fishes	13.6	10.0	85.4	80.0	1.7	1.0	19.3	11.0	85.2	80.0
Amphibians & reptiles	8.2	6.0	84.0	83.3	1.4	1.0	7.6	6.0	83.2	83.3
Birds	37.0	20.0	83.9	80.0	5.1	1.0	21.5	20.0	83.6	80.0
Rodents	36.1	18.0	83.1	83.3	3.6	1.0	31.1	15.0	83.0	81.7
Other mammals	16.2	10.0	81.1	85.7	15.9	1.0	14.1	10.0	81.7	88.0
Non-human primates	16.5	12.0	79.8	81.5	12.5	1.0	16.6	12.0	83.4	83.0
Humans	22.4	20.0	87.8	90.0	11.2	1.0	29.4	20.0	81.5	83.2

Note. Evaluation windows refer to the number of consecutive trials used to assess criterion attainment. Criteria indicate the percentage of correct responses required for success. Values summarize design parameters across keeper cells for each taxon.

Table 5

Loadings on the First Two Discriminant Dimensions for Cues and Dependent Variables

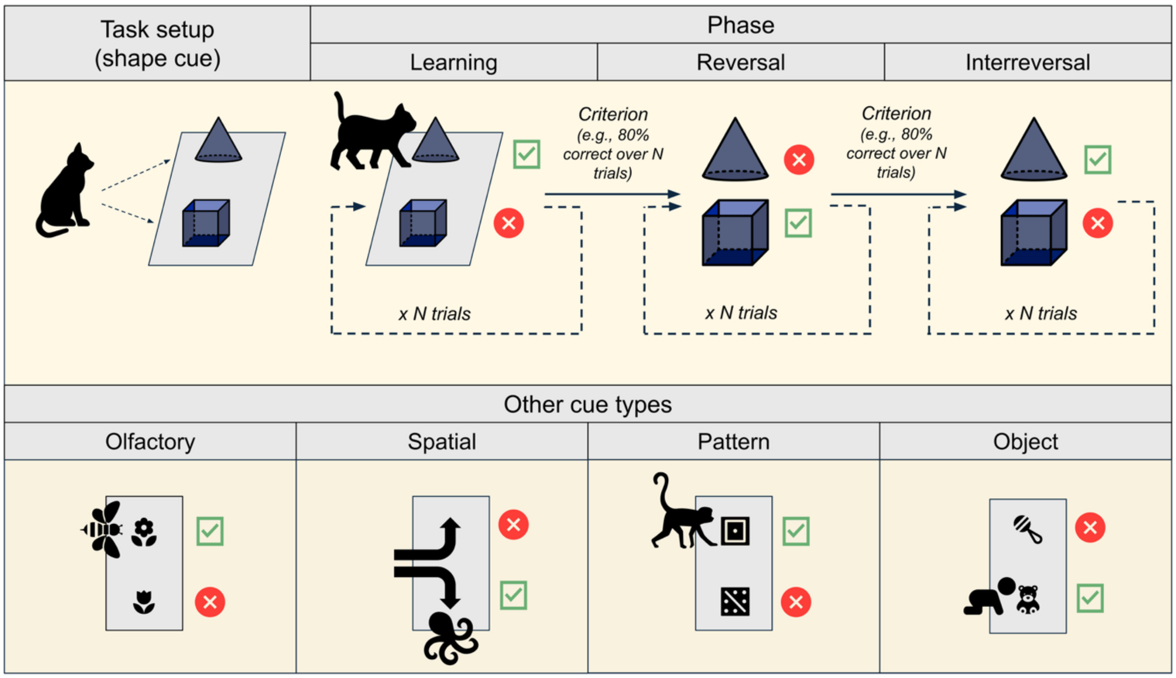
Variable	Cue LDA		Dependent variable LDA	
	LD1	LD2	LD1	LD2
	Cue loadings			
Size cues	-2.934	0.185	—	—
Shape/pattern cues	-2.095	-0.610	—	—
Color/brightness cues	0.806	1.242	—	—
Object cues	-1.618	-0.785	—	—
Spatial/position cues	0.322	-1.650	—	—
Olfactory cues	3.118	-1.887	—	—
	Dependent variable loadings			
Learning speed	—	—	2.273	0.114
Overall performance	—	—	0.841	1.998
Response speed	—	—	-0.073	1.198

Note. Loadings represent the contribution of each variable to the first two linear discriminants in each model. Blank cells indicate that the variable was not included in that model. Higher absolute loadings indicate stronger discriminatory power on the corresponding axis.

Figures

Figure 1

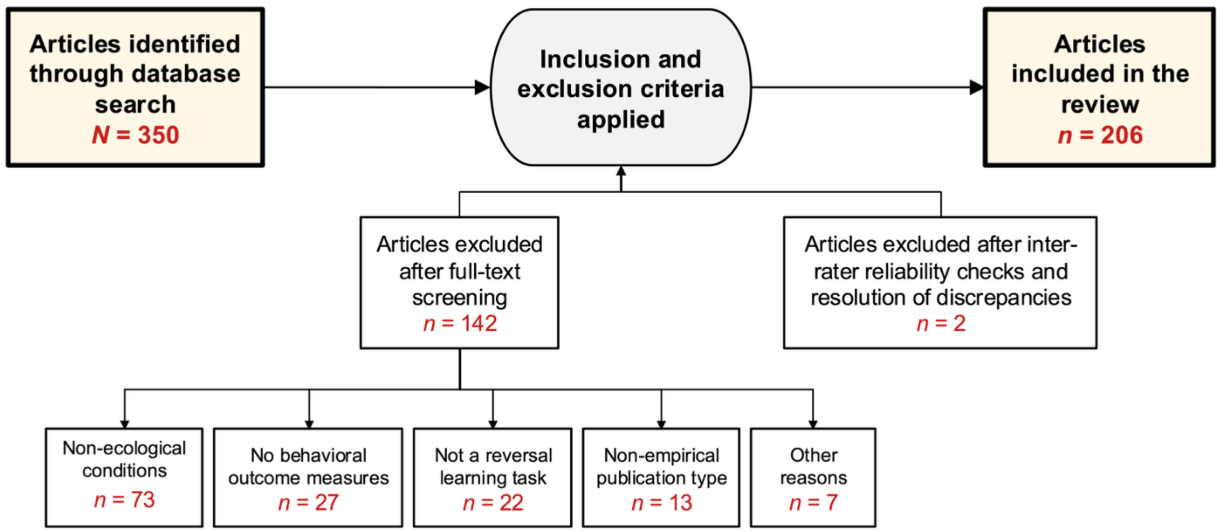
Schematic Representation of a Reversal Learning Task



Note: *Top:* A standard reversal learning task using shape cues (pyramid vs. cube). The leftmost section illustrates the task setup showing a subject presented with two stimuli on a testing apparatus. The procedure progresses through three phases: (1) Learning phase—the subject learns to associate one stimulus (e.g., pyramid) with reward over repeated trials (×N trials) until reaching criterion (e.g., 80% correct over N trials); (2) Reversal phase—reward contingencies reverse, requiring the subject to inhibit the previously learned association and acquire the new one (cube now rewarded, pyramid unrewarded); (3) Interreversal phase—in serial reversal designs, contingencies reverse again to assess learning-to-learn effects across successive reversals. Dashed arrows indicate temporal progression; dashed boxes demarcate trial blocks. Green checkmarks denote correct responses; red crosses denote incorrect responses. *Bottom*: Examples of alternative cue modalities used across taxa, illustrating methodological variation: olfactory cues (e.g., floral scents in bees), spatial cues (e.g., left vs. right in octopuses), pattern cues (e.g., visual designs in monkeys), and object identity cues (e.g., toys in human infants). This cue diversity reflects species' sensory ecology and testing constraints, contributing to methodological heterogeneity in reversal learning research.

Figure 2

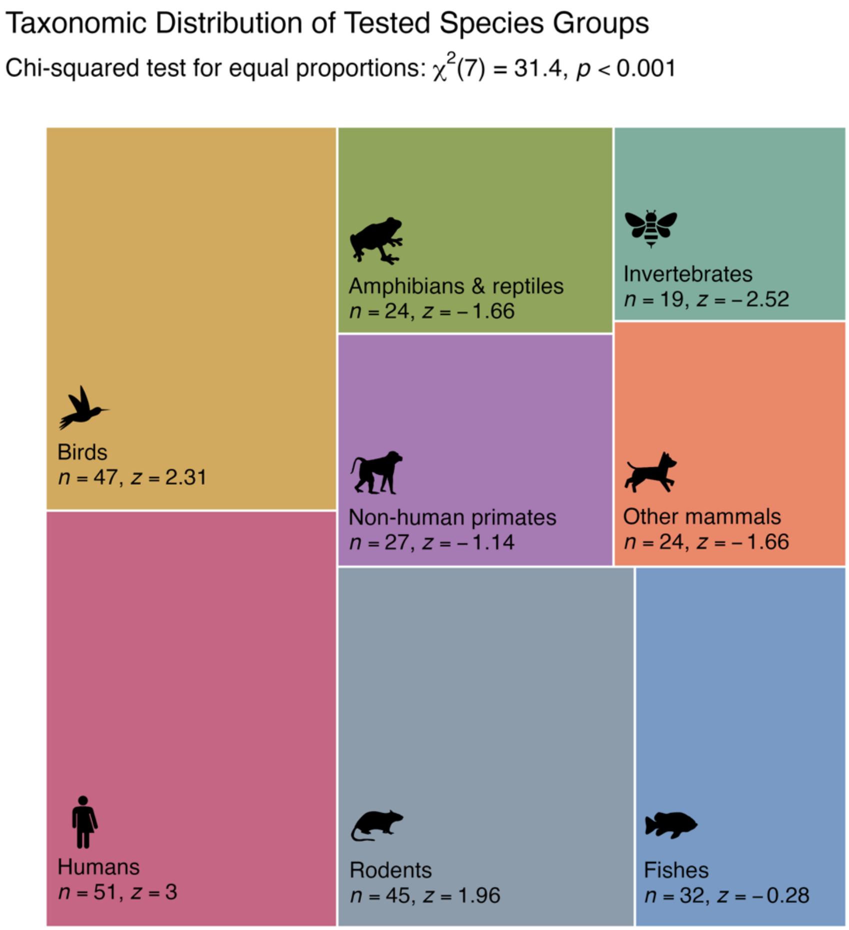
Flowchart of the Literature Search and Screening Process



Note. The diagram illustrates the progression from initial database search through final article inclusion. Yellow boxes denote entry and exit points: articles identified through database search (*N* = 350) and articles ultimately included in the review (*n* = 206). The rounded-edge box represents the decision point where inclusion and exclusion criteria were systematically applied. White boxes below reflect concrete procedural outcomes, showing the number of articles excluded at different stages and for specific reasons: 142 articles were excluded after full-text screening, with reasons including non-ecological testing conditions (n = 73), absence of behavioral outcome measures (*n* = 27), tasks not meeting reversal learning criteria (*n* = 22), non-empirical publication types (*n* = 13), and other reasons (*n* = 7). An additional 2 articles were excluded after inter-rater reliability checks and resolution of discrepancies. Arrows indicate the flow throughout the screening process.

Figure 3

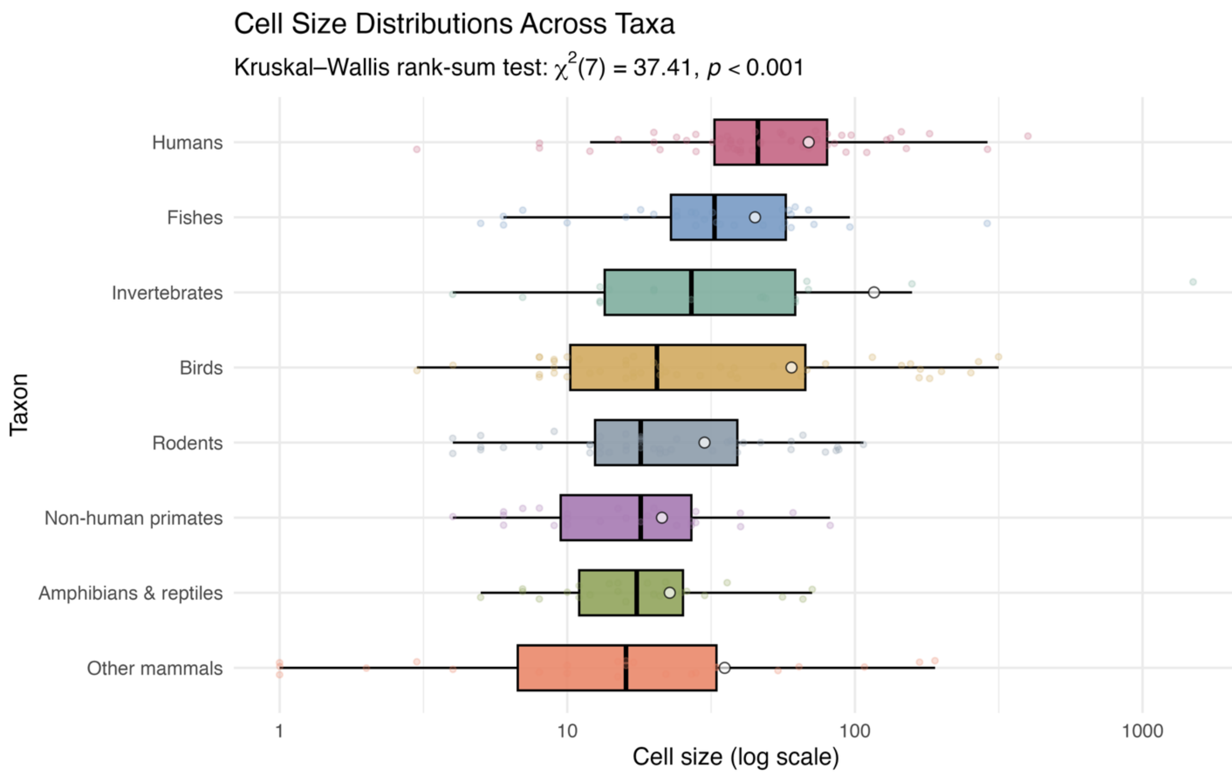
Taxonomic Distribution of Tested Species Groups (Cells)



Note. The treemap displays the relative representation of eight major taxonomic groups, with box sizes proportional to the number of cells for each group (n values). A cell is defined as a unique group of subjects from a given species tested under a specific experimental design. A chi-squared goodness-of-fit test reveals significant deviation from a uniform distribution across taxonomic groups ($\chi^2(7) = 31.4$, $p < 0.001$), indicating taxonomic bias in research representation. Standardized residuals (z-values) shown within each box quantify the degree of over- or underrepresentation relative to expected frequencies under equal distribution: positive z-values indicate overrepresentation (more cells than expected for that group), while negative z-values indicate underrepresentation (fewer cells than expected for that group).

Figure 4

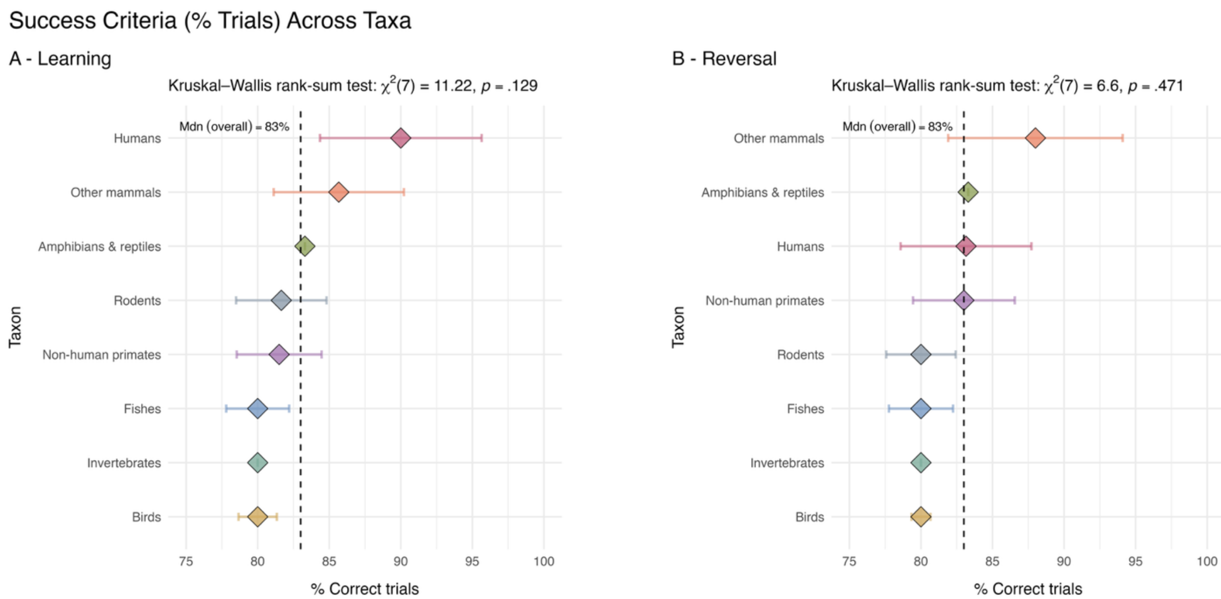
Cell Size Distributions Across Taxa



Note. Each colored point represents a cell; boxplots show the median (central vertical line) and interquartile range (box edges spanning the 25th to 75th percentiles), with whiskers extending to 1.5 times the interquartile range; white circles indicate the mean cell size for each taxon. The x-axis is displayed on a logarithmic scale to accommodate the wide range of sample sizes across cells. A Kruskal–Wallis rank-sum test reveals significant differences in cell size distributions across taxonomic groups ($\chi^2(7) = 37.41$, $p < 0.001$), indicating that sample sizes vary systematically by taxon.

Figure 5

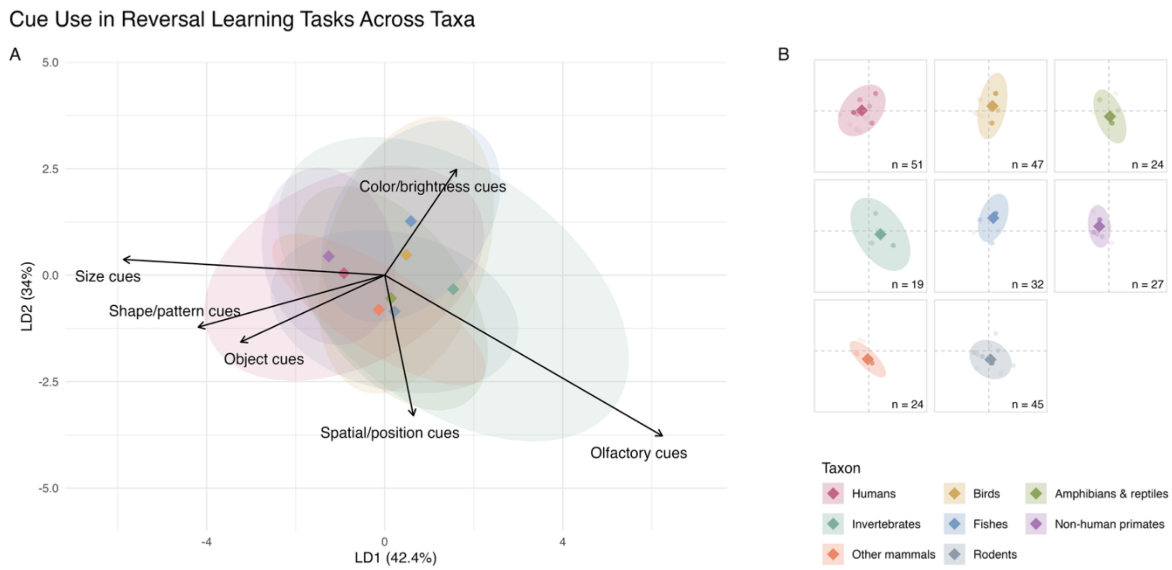
Success Criteria (% Trials) Across Taxa



Note. Panels show success criteria for learning (A) and reversal (B) phases, expressed as the percentage of correct trials required for subjects to be deemed to have learned or reversed the initial association. Diamonds denote median values by taxon, with whiskers representing robust confidence intervals. The vertical dashed line indicates the overall median across all taxonomic groups (83%).

Figure 6

Cue Use in Reversal Learning Tasks Across Taxa



Note. Panel A shows the results of a linear discriminant analysis (LDA) of cue use across taxa in reversal learning tasks. Ellipses represent 95% confidence regions around taxonomic groupings, and arrows indicate cue loadings on the first two discriminant dimensions. Panel B is a companion graphic displaying the same LDA space faceted by taxon, enabling closer examination of within-group distributions. Each point represents a study cell. Colored diamonds mark group centroids, consistent across both panels. The number of study cells contributing to each taxon is indicated in the bottom right of each facet.

S1 Table

Inter-rater reliability (IRR) measures for categorical and numeric variables

IRR for categorical variables (Gwet’s AC1)			
Pair	AC1	*SE*	Interpretation
Learning speed	0.626	0.076	Good
Overall performance	0.800	0.054	Very good
Response speed	0.824	0.051	Very good
Overtraining	0.904	0.037	Very good
Reinforcement	0.771	0.047	Good
Color and brightness cues	0.746	0.065	Good
Object cues	0.862	0.042	Very good
Olfactory cues	0.970	0.018	Very good
Shape and pattern cues	0.851	0.046	Very good
Size	0.930	0.027	Very good
Spatial location and position cues	0.757	0.063	Good
IRR for numeric variables (Cohen’s Kappa)			
Pair	Kappa	*SE*	Interpretation
Number of learning trials	0.734	0.031	Good
% correct responses to learning criterion	0.669	0.035	Good
Number of reversal trials	0.566	0.028	Moderate
Number of reversals	0.740	0.049	Good
% correct responses to reversal criterion	0.623	0.042	Good
Sample size	0.729	0.016	Good

Note: Interpretation categories for inter-rater reliability follow Altman’s (1991) thresholds: < 0.20 = Poor; 0.21–0.40 = Fair; 0.41–0.60 = Moderate; 0.61–0.80 = Good; > 0.80 = Very good. AC1 = Gwet’s Agreement Coefficient 1; SE = Standard Error. AC1 values were used for categorical variables. Kappa values were used for numeric variables.

S2 Table

Instruction manual for coders

Category	Frequently Asked Question	Response
Instructions on the exclusion criteria	How to determine that an article "focuses on neural activity rather than on behavior"?	Exclude articles that report only neural data. If behavioral data are also reported, the article is a keeper, as it provides the necessary information.
	When is a study "not focused on reversal learning under ecological conditions"?	If it examines effects of factors not typically present in the species' natural environment (e.g., drug use, surgical implants, genetic modifications). Studies involving common conditions (e.g., exercise, fear, food deprivation, day/night cycles) are keepers.
	What if the study includes multiple groups, with only some of them not tested "under ecological conditions"?	If any group is tested under non-ecological conditions (e.g., gene knockdown, psychoactive drugs), the study is a non-keeper. For instance, a placebo vs. drug group design is excluded.
Supplementary materials	Should information in supplementary material be considered for the coding?	No. Only information in (or easily inferred from) the main text should be coded.
Adding rows	When to add a new row to the spreadsheet?	Add a new row for each species tested (based on Latin name; e.g., wild vs. domestic). Do not add rows for different conditions (e.g., cue types or populations). If the paper includes multiple studies (e.g., Study 1 and Study 2), each gets its own row.
Reporting sample size	What to include under "sample size"?	Count the number of individuals per species at the start of the discrimination phase, before any dropouts. E.g., 20 dogs and 30 rats of the same species = two rows, with sample sizes 20 and 30. Two eagle populations of the same species with 10 subjects each = one row, sample size = 20.

Reinforcement type	When is the reinforcement type "Reward - material"?	When individuals receive tangible or symbolic rewards for completing the task (e.g., money, objects, points). Common in human adult studies (e.g., casino-style games).
Learning criterion	How to report the learning criterion?	Use two columns: (1) percentage of correct responses required, and (2) number of trials that percentage refers to. E.g., 16/20 = 80% out of 20 trials; 8 correct in a row = 100% out of 8; repeated 8/10 sessions = 80% out of 10. If no criterion is applied, use 0% out of X trials.
Cues	How to report that several cue types were used for one species?	There is one row per species. Mark "Yes" for each cue type used. E.g., if a study with infants uses both shape and spatial cues, mark "Yes" for shape and "Yes" for spatial cues.
	What is the difference between object cues and other cue types?	Use object cues if the reward depends on choosing between distinct objects, even if they differ in shape/size. Use shape, size, or color cues only when that is the sole parameter varying. E.g., ball vs. stick = object cue; red triangle vs. green triangle = color cue (not object cue).

S3 Table

Species Coverage by Taxon: Percentage of Species Included in the Review Relative to Total Described Species (NCBI Taxonomy, May 14, 2025)

Group (as defined in *taxon_revised*)	Terms used during the search (NCBI)	Species in review	Total species (NCBI)	Proportion covered
Invertebrates	Arachnida, Cephalopoda, Insecta	7	172666	0.004%
Fishes	Actinopterygii, Chondrichthyes, Coelacanthimorpha, Cyclostomata	13	24311	0.053%
Amphibians & reptiles	Amphibia, Squamata,	21	16130	0.130%
Birds	Aves	24	10611	0.226%
Rodents	Rodentia	9	2340	0.385%
Other mammals	Ancodonta, Perissodactyla, Ruminantia, Suina, Tylopoda; Canidae; Eupleridae; Musteloidea; Pinnipedia	11	506	2.174%
Non-human primates	Primata (not Homo sapiens)	14	490	2.857%
Humans	Homo sapiens	1	1	100.00%